

ASSIGNMENT 1 · BONG STUDY HUB · FOUNDATION BATCH 2026

Think Like a Programmer

A homework assignment for Class 01: Welcome to Computer Science

Estimated time: 30–45 minutes · Tools needed: pen and paper, or a simple document · Prerequisite: none

In class, we learned that solving a problem with a computer follows a chain, and that almost anything can be described as input, process, and output:



An **algorithm** is not the same as a **program** — an algorithm is the recipe, a program is the dish, written in a specific programming language. This assignment asks you to do the recipe part only. **You will not write any code.**

1

Your Task

Step 1 — Choose ONE everyday process from the list below (or propose your own to your instructor):

- ATM withdrawal
- Booking a cab
- Making tea
- Booking a train ticket
- Ordering food
- Online shopping
- Sending a message
- Going to college

Step 2 — Identify the Input, Process, and Output of your chosen process.

Step 3 — Write an algorithm with **at least 5 clear, numbered steps**, in plain language.

- No code. No programming syntax. Plain sentences only.
- Each step should describe one clear action.
- Steps must be in the order they actually happen.
- Write it precisely enough that someone who has never done this task could follow it without guessing.

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Beginner Example (Fully Worked)

This example shows exactly the level of detail expected. You may not copy this process — choose a different one from the list — but match this style.

Chosen process: Ordering food online

Part	Description
INPUT	Restaurant, food item, quantity, delivery address, payment method
PROCESS	Selecting a restaurant, selecting food, adding items to the cart, entering the delivery address, choosing a payment method, confirming the order
OUTPUT	Order placed successfully

Algorithm

1. Open the food ordering app.
2. Choose a restaurant.
3. Select the food items you want and choose the quantity of each.
4. Add the selected items to your cart.
5. Enter or confirm your delivery address.
6. Choose a payment method.
7. Review your order and total price.
8. Confirm the order.
9. Receive confirmation that the order has been placed.

Notice: 9 steps, one clear action each, no missing assumptions, no code.



Optional Challenge — Not Required

This section is entirely **optional**. It will not be held against you if you skip it, and it is not required to score full marks. Attempt it only if you want to push your thinking further.

If you'd like the challenge, add to your algorithm:

- At least two decisions — a point where what happens next depends on a condition.
- At least one possible failure — something that could go wrong.
- At least one assumption your algorithm quietly makes.
- At least one edge case — an unusual situation your algorithm should still handle.

You do not need to know any programming syntax (no “if/else”) — just describe the decision in plain English.

Payment successful?

YES → Order is confirmed, and a confirmation message is sent.

NO → Payment retry is offered; if it fails again, the order is not confirmed and the cart is saved for later.



Submission Format

Choose whichever is easiest for you — no GitHub, no programming software, no special formatting tools required.

- **Option A:** Handwrite your answer, then photograph or scan it.
- **Option B:** Type it in a simple document (Word, Google Docs, plain text).
- **Option C:** Submit as a simple PDF.

Use AI tools only to understand concepts or improve your thinking. Your submitted solution should reflect your own reasoning.

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Your Submission Template

Use this page directly (print and fill in, or copy the structure into your own document).

Name: _____

College: _____

Branch: _____

Date: _____

Chosen process: _____

INPUT:

PROCESS:

OUTPUT:

ALGORITHM:

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____
9. _____
10. _____

5

Optional Challenge (space for your answer)

Leave blank if you're not attempting the challenge — that's completely fine.

✓

Before You Submit — Self-Check

- ☐ I identified the input.
- ☐ I identified the process.
- ☐ I identified the output.
- ☐ I wrote at least 5 steps.
- ☐ My steps are in logical order.
- ☐ Someone else could understand my steps.
- ☐ I did not use code.
- ☐ I reviewed my work before submitting.